MMT: Mixed-Mask Transformer for Remote Sensing Image Semantic Segmentation

Zhe Xu, Jie Geng, *Member, IEEE*, and Wen Jiang

***Abstract*— Remote sensing image semantic segmentation is a crucial step in the intelligent interpretation of remote sensing. Most of the current approaches are based on the attention mechanism to enhance long-range representations. However, these works ignore the key problem of foreground–background imbalance, and their performances encounter a bottleneck. In this article, we introduce mask classification into remote sensing image interpretation for the first time and propose a novel mixed-mask Transformer (MMT) for remote sensing image semantic segmentation. Specifically, we propose a mixed-mask attention mechanism, a simple but effective module, which assists the network to learn more explicit intraclass and interclass correlations by capturing long-range interdependent representations. In addition, a progressive multiscale learning strategy (MSL) is proposed to solve the problem of large-scale-varied targets in remote sensing images, which integrates semantic and visual representations of different scale targets by efficiently utilizing large-scale feature maps in Transformer. Experimental results show that the proposed MMT exceeds the existing alternative approaches and achieves state-of-the-art performance on three semantic segmentation datasets.**

***Index Terms*— Attention mechanism, foreground–background imbalance, remote sensing, semantic segmentation, Transformer.**

I. Introduction

REMOTE sensing image semantic segmentation is a crucial step in the intelligent interpretation of remote sensing, which aims to assign a semantic label to each pixel for a given image. With the development of deep learning technologies, remote sensing image semantic segmentation is playing an increasingly vital role in urban planning [1], autonomous driving [2], environmental monitoring [3], and even military fields.

In recent years, a convolutional neural network (CNN) [4] has been demonstrated to be effective in several computer vision tasks, such as classification, detection, and segmentation, relying on its ability to extract more discriminative and robust features in the deep networks [5]. For semantic segmentation tasks, a fully convolutional network (FCN) [6] is proposed to implement end-to-end prepixel dense prediction directly through convolution layers and upsampling layers, and has achieved breakthroughs in natural scenes. Due to the simple framework and outstanding performance of FCN, many studies [7], [8], [9], [10] have made pioneering progress in applying it to remote sensing images. However, different from natural scene images, remote sensing images are abundant with spatial information, especially long-range contextual interactions.

In order to effectively capture long-range spatial information, various attention mechanisms have been applied to remote sensing images, which promotes the rapid development of remote sensing image semantic segmentation. For example, Panboonyuen et al. [11], Geng et al. [12], and Chen et al. [13] explored spatialwise and channelwise attention modules to enhance global feature representation, and SBANet [14] proposed a boundary attention module that further aggregates hierarchical boundary features and reaches the state of the art. Furthermore, a network architecture mainly consisting of attention modules, Transformer, was proposed by Dosovitskiy et al. [15] and rapidly advanced in the field of computer vision. Xu et al. [16] and Zhang et al. [17] introduced Transformer into remote sensing images and achieved remarkable performance improvements by adequately capturing contextual interactions. Li et al. [18], [19] studied remote sensing image semantic segmentation tasks from the perspective of cross-domain adaptive and ontological reasoning, respectively. The attention-based method has been demonstrated to have a powerful modeling capability for long-range dependencies and can learn the global spatial representation of remote sensing images. However, redundant background features in global spatial information become the bottleneck of its performance improvement. It is time to focus on another crucial problem in high-resolution remote sensing images, which has been neglected in previous studies, namely, foreground–background imbalance. Many studies [20], [21], [22] clearly pointed out that the foreground ratio of remote sensing images is much less than that in natural images, and relevant studies have been conducted on the foreground object segmentation task. As for the semantic segmentation task of remote sensing image land cover, the foreground ratio varies among different classes, and the foreground–background imbalance problem is more challenging.

Inspired by MaskFormer [23], which shows the segmentation task can be formulated to predict a set of binary masks, each associated with a single class prediction, we observe that mask classification naturally separates foreground and

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The authors are with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an 710072, China (e-mail: gengjie@nwpu.edu.cn).

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background to indicate their proportions explicitly. However, MaskFormer only focuses on the foreground completely ignoring the background, while the interactive information between foreground and background is essential for remote sensing images. How to capture context-interdependent representations within a foreground–background balance way while preserving the spatial long-range information of remote sensing images is the key problem.

In this article, we introduce mask classification into remote sensing image interpretation for the first time and propose a novel mixed-mask Transformer (MMT) for remote sensing image semantic segmentation. MMT aims to explore a new paradigm for remote sensing image semantic segmentation tasks within a foreground–background balance way. Specifically, first, our MMT introduces the mask classification pipeline to separate the foreground and background of remote sensing images using prediction masks and, second, proposes the mixed-mask attention (MMA) mechanism by using a random mask for the foreground region and grid mask for the background region, which not only eliminates redundant background features but also ensures that the long-range contextual exploitation is not affected. In addition, the proposed efficient MSL can exploit global long-range representation with linear complexity. Overall, the main contributions of our work can be summarized as follows.

1) We introduce mask classification into remote sensing image interpretation for the first time and propose a novel MMT to assist the network in learning more explicit long-range interdependent representations, thus developing a new paradigm for remote sensing image semantic segmentation within a foreground–background balance way.
2) To solve the problem of large-scale-varied targets in remote sensing images, a progressive MSL is proposed to efficiently utilize large-scale feature maps in Transformer.
3) Experimental results show that the proposed MMT exceeds the existing alternative approaches and achieves state-of-the-art performance on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam 2-D semantic labeling contest datasets.

The rest of this article is organized as follows. Section II presents the related work. The proposed MMT for remote sensing image semantic segmentation is described in Section III. In Section IV, the performance of the proposed method is analyzed in the experiments. Conclusions are finally summarized in Section V.

II. Related Work

In this section, we review some recent works on semantic segmentation of natural scenes and remote sensing images. Then, we present mask classification-based approaches that are more relevant to our work.

A. General Semantic Segmentation

Semantic segmentation is to segment the image into per-pixel regions with semantic information and identify the semantic classes of each region to achieve semantic inference from the bottom level to the top level. FCN-based [6] methods leverage FCNs to implement prepixel dense prediction of any desired resolution, which has made a great breakthrough in semantic segmentation. This idea gradually evolved into the encoder–decoder structure with several variations. Concretely, SegNet [24] records pooling indices to preserve the contour information of the image, thus performing nonlinear upsampling and reducing the number of parameters. UNet [25] introduces a U-shaped structure that can capture contextual information and precisely locate segmentation boundaries to improve performance and robustness. DeepLab series [26], [27], [28] and DeepLabv3+ [29] employ ASPP to capture contextual information of different scales and improve the decoder to recover the original pixel information. In addition, DFN [30], CCNet [31], SANet [32], SPGNet [33], and RANet [34] adopt various attention mechanisms to further exploit the global spatial modeling capability and capture long-range dependency representations, which yields outstanding semantic segmentation performance.

B. Remote Sensing Image Semantic Segmentation

Deep learning methods have made remarkable progress in remote sensing image semantic segmentation. For example, Nogueira et al. [35], Li et al. [36], and Wu et al. [37] directly apply the general semantic segmentation method to remote sensing images and achieve decent results. Considering that remote sensing images have more abundant spatial information than natural scene images, many studies have attempted to design various deep networks suitable for remote sensing images. Xu et al. [38] proposed a high-resolution context extraction network to keep the spatial information and integrate the low-to-high features. Liu et al. [39] developed an adaptive fusion network to adaptively focus on size-varied objects on multilevel feature maps. Yang et al. [40] adopted a multipath encoder and attention-fused structure to perform multipath and multilevel feature fusion simultaneously. Li et al. [41] proposed a multiattention mechanism with linear complexity to aggregate relevant contextual features hierarchically. CCANet [42] proposed a coarse-to-fine attentional network to obtain explicit long-range context information by constraining class information. Existing methods have made significant progress mainly through enhancing long-range spatial representations, but their performance has encountered a bottleneck. This article focuses on the problem of foreground–background imbalance in remote sensing images and breaks the performance of semantic segmentation by enhancing the contextual interdependencies between foreground and background.

C. Mask Classification

Mask classification has been widely applied to the instance-level segmentation task. Mask R-CNN [43] classifies the proposals generated by RPN and predicts a series of masks for each segment to achieve instance segmentation. SOLO [44] splits the image into several regions via category- and instance-aware modules, and generates instance masks for objects

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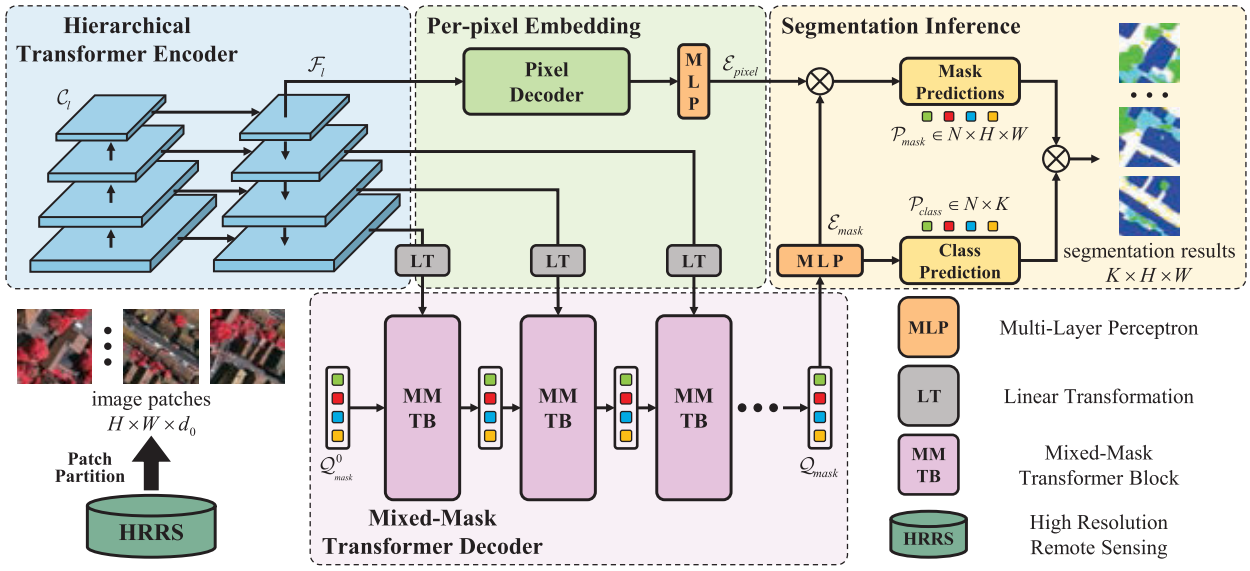


Fig. 1. Framework of the proposed MMT for remote sensing image semantic segmentation.

in each region. These tasks require determining the region of the segment target before generating the mask, making the per-pixel semantic segmentation task challenging. DETR [45] proposes the set prediction mechanism, which yields a single class prediction per segment by learning object queries, which is an end-to-end semantic segmentation based on the probability of mask. Recently, MaskFormer [23] and Mask2Former [46] attempt to formulate the segmentation task as predicting a set of binary masks, each associated with a single class prediction, to obtain a unified segmentation framework. Using the mask classification pipeline proposed in MaskFormer, our approach divides each mask into foreground and background, and proposes a mixed-mask mechanism for remote sensing image semantic segmentation.

III. Methodology

In this section, an end-to-end MMT is proposed for remote sensing semantic segmentation, whose framework is shown in Fig. 1. We first introduce the motivation and the pipeline of mask classification that MMT is built on. Then, the proposed modules of MMT are described in detail. Finally, we summarize MMT and discuss the reasons for its effectiveness.

A. Mask Classification Pipeline

Traditional approaches typically formulate general semantic segmentation as a per-pixel classification task, and so far, the methods of remote sensing image semantic segmentation are also in the same way. The method based on per-pixel classification has shown good performance in the field of remote sensing, but it is still difficult to deal with a crucial problem of high-resolution remote sensing images, which is foreground–background imbalance. Inspired by MaskFormer [23], we observe that a segmentation task can be formulated to predict a set of binary masks, each associated with a single class prediction, which naturally separates foreground and background to indicate their proportions explicitly. However, MaskFormer only focuses on the foreground completely ignoring the background, while the interactive information between foreground and background is essential for remote sensing images. To address the problem of foreground–background imbalance head-on, we develop mask classification and propose a novel MMA mechanism to learn the long-range interdependencies between foreground and background in a balanced way. In addition, in order to solve the problem of large-scale-varied targets in high-resolution remote sensing images, we utilize the feature pyramid structure (FPN) [47] for the Transformer encoder and propose an MSL for the Transformer decoder to integrate semantic and visual representations into different scale targets efficiently.

A standard mask classification pipeline consists of three components. First, a backbone network extracts depth features $\mathcal{F}$ from images. In order to enable the Transformer decoder to achieve multiscale learning, a hierarchical Transformer encoder is introduced as the backbone to extract multiscale feature maps. Second, a pixel decoder upsamples low-resolution features from the output of the backbone to generate high-resolution per-pixel embeddings $\mathcal{E}_{\text{pixel}}$. Third, a Transformer decoder trains learnable mask queries $\mathcal{Q}_{\text{mask}}$ via learning from multiscale features and learnable mask queries. Note that the proposed MMA mechanism and MSL are both key innovations of the Transformer decoder. In the semantic segmentation inference, mask queries $\mathcal{Q}_{\text{mask}}$ directly generate the class predictions $\mathcal{P}_{\text{class}}$ with its corresponding mask embeddings $\mathcal{E}_{\text{mask}}$ via MLP, while the binary mask predictions $\mathcal{P}_{\text{mask}}$ are obtained via a cross-product between per-pixel embeddings $\mathcal{E}_{\text{pixel}}$ and mask embeddings $\mathcal{E}_{\text{mask}}$. Finally, the final semantic segmentation result is obtained by combining mask predictions $\mathcal{P}_{\text{mask}}$ with class predictions $\mathcal{P}_{\text{class}}$ using matrix multiplication.

B. Hierarchical Transformer Encoder

The hierarchical Transformer encoder is the backbone module of the MMT. It extracts feature maps of different scales for further multiscale learning of the Transformer decoder. Since ViT [15] maintains the same number of tokens across the entire network, which has fixed the image resolution,

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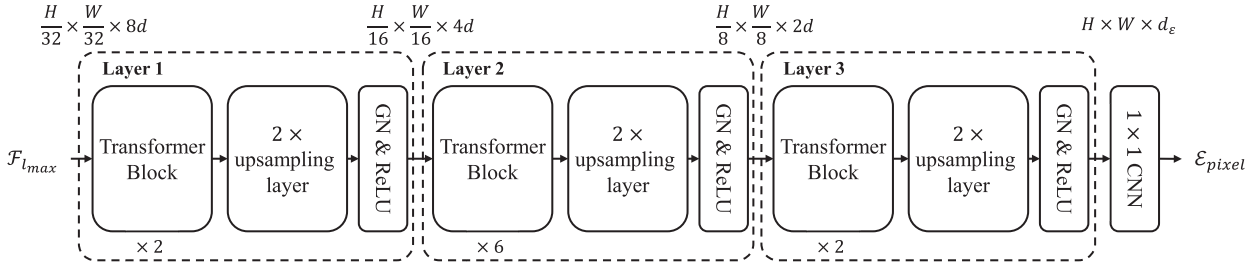


Fig. 2. Framework of the pixel decoder.

we introduce the patch merging layer to build multiscale feature maps according to [48]. In detail, given a remote sensing image of $H \times W \times d_0$ size, multiscale feature maps are generated by hierarchical Transformer encoder as $\mathbf{C} = \{\mathcal{C}_l \mid \mathcal{C}_l \in \mathbb{R}^{d_l \times H_l \times W_l}, l \in [l_{\min}, l_{\max}]\}$, where l is the level of the encoder, and $H_l = H/2^l$, $W_l = W/2^l$, and d_l denote the height, width, and channel dimension of the output feature maps, respectively. Different levels of feature maps have different scales, and the higher the level, the lower the resolution of feature maps.

Furthermore, in order to enhance the representation of small targets, multiscale features are preliminarily fused by FPN [47]. The high-level features have a larger receptive field and contain abundant semantic information, while the low-level features have a larger resolution and can provide a visual representation of small targets. The FPN integrates the complementary information of high- and low-level features to obtain more discriminative features $\mathbf{F} = \{\mathcal{F}_l \mid \mathcal{F}_l \in \mathbb{R}^{d_l \times H_l \times W_l}, l \in [l_{\min}, l_{\max}]\}$, where F_l is defined as

$$\mathcal{F}_l = \begin{cases} f(\text{conv}(\mathcal{C}_l)), & l = l_{\max} \\ f(\text{upsample}(\mathcal{F}_{l+1}) + \mathcal{C}_l), & \text{otherwise} \end{cases} \tag{1}$$

where $f(\cdot)$ represents the smooth feature function in [47], conv represents the 1×1 convolution function, and upsample represents the bilinear upsampling function.

It should be noted that, for the l feature maps of different scales generated by the hierarchical Transformer encoder, the highest level feature map with the lowest resolution is input into the pixel decoder to generate per-pixel embeddings, while the other feature maps are input into the MMT decoder (MTD) for multiscale learning.

C. Pixel Decoder

The pixel decoder is predominantly used to generate per-pixel feature embeddings, which contain per-pixel global representation. As shown in Fig. 2, the pixel decoder consists of three Transformer layers, and each layer includes a Transformer block, an upsampling layer, a GroupNorm layer, and the ReLU activation. After the third Transformer layer, there is also a single 1×1 convolution layer to project the output features to the corresponding channel dimensions. The pixel decoder takes the highest level feature maps $F_{l_{\max}}$ as input and continuously upsamples the feature maps to obtain per-pixel embeddings $\mathcal{E}_{\text{pixel}} \in \mathbb{R}^{d_\mathcal{E} \times H \times W}$, where $d_\mathcal{E}$ denotes the channel dimension of embeddings. For the general per-pixel classification-based segmentation model, the pixel decoder is used to directly output the final result, which has the problem of foreground–background imbalance due to the sparse distribution of targets in remote sensing images. In our observation, per-pixel embedding contains rich global context information and serious interclass similarity and intraclass inconsistency. Our mask classification method extracts mask embeddings and integrates them with per-pixel embeddings, which can reduce the foreground–background imbalance and enhance the interaction of long-range representations.

D. Mixed-Mask Transformer Decoder

MTD is the key module of the proposed MMT, which includes multilevel Transformer blocks to handle multiscale features from hierarchical Transformer encoder. To solve the pain point of foreground–background imbalance in remote sensing image semantic segmentation, we designed a novel MMA mechanism for the Transformer decoder, which constrains cross-attention (CA) to the foreground and background regions in a balanced way. In addition, in order to solve the problem of large-scale-varied targets in high-resolution remote sensing images, we propose an efficient MSL for the Transformer decoder to integrate semantic and visual representations between large and small targets efficiently. We now introduce these improvements in detail.

1) Architecture of Mixed-Mask Transformer Decoder: As shown in Fig. 3(a), a standard Transformer decoder (STD) mainly consists of a multihead self-attention (SA) layer, a CA layer, and a multilayer perceptron (MLP) layer, each of which employs residual connections between the sublayers and is followed by layer normalization. The SA layer takes learnable query features as input, while the CA layer performs attention operations over the output feature sequences of the Transformer encoder. The CA layer captures the global context of feature maps, which is the main reason for the slow convergence of Transformer-based models and foreground–background imbalance.

In order to better match the proposed MMA mechanism, we adjust the architecture of the STD. As shown in Fig. 3(b), we switch the order of the SA layer and the CA layer, and replace the CA layer with the proposed MMA layer. Specifically, a mask queries $\mathcal{Q}^l_{\text{mask}}$ serves as the query of the MMA layer, and $\mathcal{K}_l$ and $\mathcal{V}_l$ are obtained by the transformation of a level of feature maps $\mathcal{F}_l$, which are served as the key and value of the MMA layer, respectively. Meanwhile, attention mask $\mathcal{M}_l$ is used to constrain the area of attention operation. Details of the MMA mechanism will be covered in Section III-D2. The output of the MMT block is the new mask queries $\mathcal{Q}^{l+1}_{\text{mask}}$, which serves as the query of the next block. The adjustment of the STD masks part of the position area

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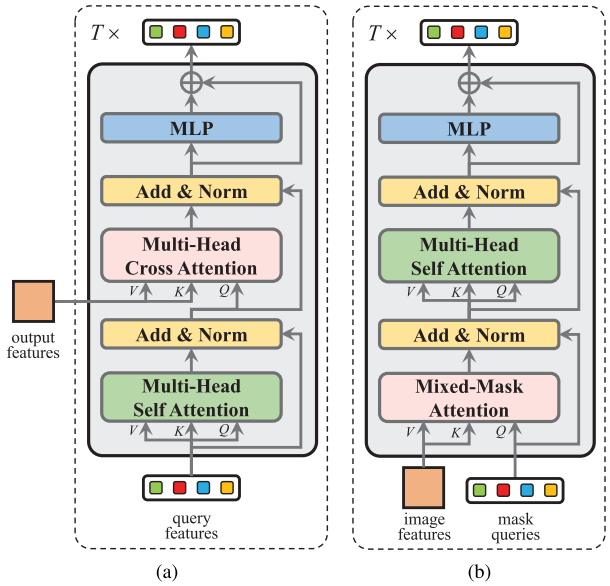


Fig. 3. Block structure comparison between (a) STD and (b) MTD.

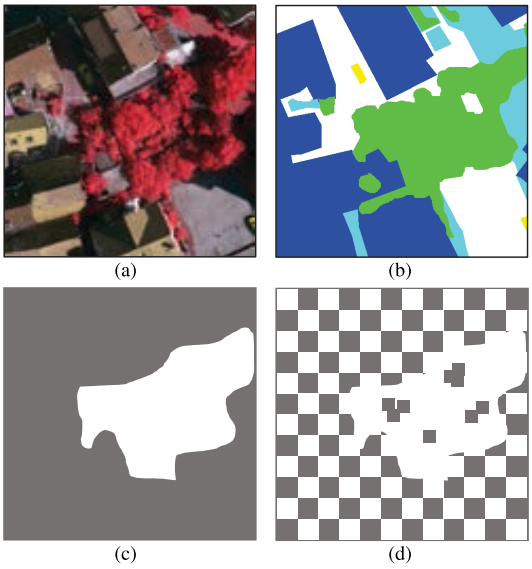


Fig. 4. Illustration of MMA mechanism. (a) IRRG image patch. (b) Ground truth. (c) Mask attention. (d) MMA.

before SA operation, which can reduce the calculation of SA and CA.

2) Mixed-Mask Attention Mechanism: As mentioned before, there is the problem of foreground–background imbalance due to the sparse distribution of targets in high-resolution remote sensing images, which leads to the bottleneck of the performance of remote sensing image semantic segmentation. MaskFormer separates foreground and background to enhance the representation of localized target regions but ignores the background information and the interaction of long-range representation, which is important for remote sensing images. To this end, we propose a novel MMA mechanism to learn the interdependencies between foreground and background in a balanced way.

As shown in Fig. 4, for the tree class, the mask attention mechanism only constrains to the foreground region of the predicted mask, while the MMA mechanism uses a mixed sampling-masking strategy to generate the new operation region based on the mask attention. Specifically, the MMA mechanism uses grid sampling in the background region, which keeps one of every two patches and uses random masking in the foreground region, which generates mask patches randomly. Background information usually has discrete distribution and local similarity, so grid sampling can maximize the utilization of background information while increasing the limited computation. As for the foreground target, recent studies [49] have shown that random masking can reduce network parameters and increase the robustness of target samples. The mixed sampling-masking strategy can enhance the long-range representation between foreground and background, and effectively solve the problem of the foreground–background imbalance in remote sensing images.

Standard CA function with residual connection is calculated as follows:

$$\mathcal{X}_l = \text{softmax}\left(\mathcal{Q}_l \mathcal{K}_l^{\mathrm{T}}\right)\mathcal{V}_l + \mathcal{X}_{l-1} \tag{2}$$

where l stands for the level of decoder and $\mathcal{X}_l \in \mathbb{R}^{N_l \times d}$ denotes query features of the lth layer of the Transformer decoder. Here, N_l denotes the number of query features, and $N_l = H_l W_l$. $\mathcal{Q}_l \in \mathbb{R}^{N_l \times d}$ denotes the query of query features $\mathcal{X}_{l-1}$ under linear transformation, and $\mathcal{K}_l, \mathcal{V}_l \in \mathbb{R}^{H_l W_l \times d}$ denote the key and value of feature map $\mathcal{F}_l$ under linear transformation.

According to Cheng et al. [23], mask attention and our mixed-masked attention both compute the attention matrix as follows:

$$\mathcal{X}_l = \text{softmax}\left(\mathcal{M}_{l-1} + \mathcal{Q}^l_{\text{mask}} \mathcal{K}_l^{\mathrm{T}}\right)\mathcal{V}_l + \mathcal{X}_{l-1}. \tag{3}$$

Different from the standard CA, the mask query $\mathcal{Q}^l_{\text{mask}} \in \mathbb{R}^{N_q \times d}$ is initiated to a learnable vector, where the number of mask queries N_q is constant. In addition, mask attention introduces attention mask $\mathcal{M}_{l-1} \in \mathbb{R}^{N_q \times H_l W_l}$ to constrain the area of attention operation. The attention mask $\mathcal{M}_{l-1}$ at feature location (x, y) is defined as

$$\mathcal{M}_{l-1}(x, y) = \begin{cases} 0, & \text{if } \text{Mask}^{l-1}_{mixed}(x, y) = 1 \\ -\infty, & \text{otherwise} \end{cases} \tag{4}$$

where $\text{Mask}^{l-1}_{mixed} \in \{0, 1\}^{N \times H_l W_l}$ is the binarized mixed-mask, which is calculated by the mask prediction of the previous $(l-1)$th layer, the background grid mask (BGM), and the foreground random mask (FRM). Mixed-mask $\text{Mask}^{l-1}_{mixed}$ is expressed as

$$\text{Mask}^{l-1}_{mixed} = \text{Mask}^{l-1}_{\text{pred}} + \text{Mask}^{l-1}_{\text{grid}} - \text{Mask}^{l-1}_{\text{random}} \tag{5}$$

where $+$ stands for the logic **OR** operation and $-$ stands for the logic **AND** operation for the binarized bool mask. Here, $\text{Mask}^{l-1}_{\text{pred}}$ denotes the mask prediction of the $(l-1)$th layer, $\text{Mask}^{l-1}_{\text{grid}}$ denotes the BGM, and $\text{Mask}^{l-1}_{\text{random}}$ denotes the FRM. The mask proportions of BGM $\text{Mask}^{l-1}_{\text{grid}}$ are determined by the size of the mask grid s_{grid}, while the mask proportions of FRM $\text{Mask}^{l-1}_{\text{random}}$ are determined by the random ratio r_{random}. The influences of these two hyperparameters on the model performance are analyzed in the subsequent experiments.

The MTD block output yields a new mask query $\mathcal{Q}^{l+1}_{\text{mask}} \in \mathbb{R}^{N_q \times d}$, which serves as the input query to the next block, in addition to a series of calculations to obtain the mask

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Algorithm 1 MSL

Input: Multi-scale feature maps $\mathbf{F} = \{\mathcal{F}_l \in \mathbb{R}^{d_l \times H_l \times W_l}\}$, per-pixel embeddings $\mathcal{E}_{pixel} \in \mathbb{R}^{d_\mathcal{E} \times H \times W}$, maximum times of Transformer decoder loops T.

Output: Mask query $\mathcal{Q}_{mask}$

Initialize $\mathcal{Q}_{mask}$ as a learnable vector.
Calculate the mask embeddings $\mathcal{E}^0_{mask}$ and the mask predictions $\mathcal{P}^0_{mask}$ by Eq. (6).
Booleanized the mask predictions $\mathcal{P}^0_{mask}$ with a threshold of 0.5 to obtain the Mask^0_{pred}.
Calculate the mixed-mask Mask^0_{mixed} by Eq. (5).
Calculate the attention mask $\mathcal{M}_0$ by Eq. (4).
repeat
  $t \leftarrow t + 1$
  for $\mathcal{F}_l$ in **F do**
    The key $\mathcal{K}_l$ and the value $\mathcal{V}_l$ are obtained by linear transformation of $\mathcal{F}_l$.
    Update the mask query $\mathcal{Q}^l_{mask}$ with Eq. (3).
    Update the attention mask $\mathcal{M}_{l-1}$ with Eq. (4), (5) and (6).
  end for
until $t \geq T$
The final mask query $\mathcal{Q}_{mask}$ is used for semantic segmentation inference through the mask classification pipeline.

prediction $\mathcal{P}_{\text{mask}}$, thus updating the binarized mixed-mask Mask^l_{mixed}. Specifically, the mask query $\mathcal{Q}^{l+1}_{\text{mask}}$ can transform the channel dimension d to the embedding dimension $d_\mathcal{E}$ by MLP, so as to obtain the mask embeddings $\mathcal{E}_{\text{mask}} \in \mathbb{R}^{N_q \times d_\mathcal{E}}$. Then, the mask predictions $\mathcal{P}_{\text{mask}} \in \mathbb{R}^{N_q \times H \times W}$ are obtained via a cross-product between the mask embeddings $\mathcal{E}_{\text{mask}}$ and the per-pixel embeddings $\mathcal{E}_{\text{pixel}}$ as

$$\begin{aligned}\mathcal{P}_{\text{mask}} &= \mathcal{E}_{\text{mask}} \otimes \mathcal{E}_{\text{pixel}} \\ &= \text{MLP}\left(\mathcal{Q}^{l+1}_{\text{mask}}\right) \otimes \mathcal{E}_{\text{pixel}}. \end{aligned} \tag{6}$$

Finally, the mask predictions $\mathcal{P}_{\text{mask}}$ are resized to the same solution of $\mathcal{F}_{l+1}$ and booleanized with a threshold of 0.5 to update $\text{Mask}^l_{\text{pred}}$.

3) Multiscale Learning Strategy: There is another problem with large-scale-varied targets in high-resolution remote sensing images. As mentioned above, we use FPN [47] in the hierarchical Transformer encoder for a preliminary fusion of multiscale features so that the feature maps of different scales both are abundant with semantic information. It is far from enough to perform FPN only in feature extraction, and we need to seek a more progressive MSL to model contextual long-range dependencies.

Introducing large-scale features in STD improves the performance but costs the quadratic computational complexity. We observe that the computational complexity of the standard Transformer mainly comes from the computation of SA and CA, where the number of query features N_l depends on the size of the input features, which is the reason for the heavy computational complexity. Moreover, if considering the features of different scales, a resize operation is required to match the size, which introduces additional computations. Fortunately, our MTD switches the order of the SA layer and the CA layer, and replaces the standard query $\mathcal{Q}_l$ with the fixed number of mask query $\mathcal{Q}^l_{\text{mask}}$ so that the output size of the Transformer decoder is fixed and not varied with the scale of the input feature maps. Therefore, a cyclic shift MSL is proposed to elegantly learn large-scale feature maps in the Transformer decoder with acceptable computational complexity. Specifically, feature maps of different scales $\mathcal{F}_l$ are input into the MTD in the cyclic shift manner, and the mask query $\mathcal{Q}^l_{\text{mask}}$ is continuously learned and updated as a learnable vector, while the attention mask $\mathcal{M}_{l-1}$ is updated to obtain finer segmentation predictions. To better understand the workflow of the proposed MSL, we further put an algorithm with detailed parameters in Algorithm 1.

For a feature map with a size of $H \times W \times d$, the computational complexity of the STD considers only the SA layer and the CA layer, while the MTD considers only the MMA layer and the SA layer. For the STD, the computation of the SA layer consists of three parts, namely, the computation of query, key, and value generation; the computation of attention; and the computation of projection. The computation of the CA layer is similar, except that the first part includes only the computation of key and value generation. The MTD switches the order of the SA layer and the CA layer, and replaces the CA layer with the proposed MMA layer. The computational complexity of the STD and MTD can be written as follows:

$$\begin{aligned}\Omega(STD) &= \Omega(SA) + \Omega(CA) \\ &= \left[3HWd^2 + 2HWN_ld + N_ld^2\right] \\ &\quad + \left[2N_ld^2 + 2N_l^2d + N_ld^2\right] \\ &= 3HWd^2 + 4N_ld^2 + 2HWN_ld + 2N_l^2d \\ &= 7HWd^2 + 4(HW)^2d \end{aligned} \tag{7}$$

$$\begin{aligned}\Omega(MTD) &= \Omega(MMA) + \Omega(SA) \\ &= \left[2HWd^2 + 2HWN_qd + N_qd^2\right] \\ &\quad + \left[3N_qd^2 + 2N_q^2d + N_qd^2\right] \\ &= 2HWd^2 + 5N_qd^2 + 2HWN_qd + 2N_q^2d \\ &= \left(5N_q + 2HW\right)d^2 + 2\left(N_q^2 + N_qHW\right)d \end{aligned} \tag{8}$$

where $N_l = HW$ is the number of query features and N_q is the number of mask queries, which is a constant much less than HW. Since N_q is fixed, the complexity of MTD is linear, while the complexity of STD is quadratic.

IV. Experiments

A. Dataset Description

To evaluate the performance of our proposed method, ISPRS Vaihingen and Potsdam 2-D semantic labeling contest datasets [50] and the LoveDA land-cover dataset [51] are used for remote sensing image semantic segmentation. ISPRS datasets consist of the true orthophoto (TOP) tiles and the corresponding digital surface model (DSM) tiles. For fairness of comparison, DSM data are not used in all experiments.

1) Potsdam Dataset: It consists of 38 tile images with infrared-red-green (IRRG) channels, whose size is 6000 × 6000 pixels. The dataset has a ground sampling distance (GSD) of 9 cm, including six typical land-cover classes,

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namely, impervious surfaces (white), buildings (blue), low vegetation (cyan), trees (green), cars (yellow), and clutter/background (red), of which clutter/background class is ignored because of its complex structures. Following the settings of benchmark organizers [50], 14 public ground-truth tiles are used as the test set, 19 tiles are used as the train set, and the remaining five tiles (IDs: 7_7, 7_8, 7_9, 7_11, and 7_12) are used for validation.

2) Vaihingen Dataset: It consists of 33 tile images with red-green-blue (RGB) channels, whose size is around 2500 × 2000 pixels. The dataset has the same typical classes as the Potsdam dataset. Among them, there are 17 public ground-truth tiles in the test set, 11 tiles are used for training, and the remaining five tiles (IDs: 11, 15, 28, 30, and 34) are used for validation.

3) LoveDA Dataset: It contains 5987 images of size 1024 × 1024 with RGB channels. The dataset contains seven land-cover classes, namely, background, building, road, water, barren, forest, and agriculture. Following the benchmark [51], 2522 images are utilized for training, 1796 images are utilized for testing, and the remaining 1669 images are utilized for validation.

B. Experimental Settings

1) Implementation Details: The proposed model is implemented in the PyTorch framework, and all experiments are conducted on four NVIDIA RTX 2080 Tis. Following [14], the input images are cropped into patches of 512 × 512 pixels with stride size of 256 pixels in the training phase and cropped into patches of the same size with the nonoverlapping manner in validation and testing phase. For data augmentation, we apply common random cropping and random flipping during each iteration of the training phase. We take MaskFormer [23] as our baseline model and the SwinTransformer-Base (Swin-B) model with FPN as our hierarchical Transformer encoder (i.e., our backbone). We also replace the backbone network with ResNet-50 [52] for further comparison with other models. The backbone networks in the proposed method are all pretrained on the ImageNet-21K dataset. In addition, the maximum times of the Transformer decoder loop T are set to 9, and we use the AdamW [53] optimizer with the initial learning rate of 10^{-4} and the weight decay of 0.05; our network is trained total 10^4 epochs to obtain the optimal model for evaluation on the test set.

2) Evaluation Metrics: Following the ISPRS benchmark [50], we evaluate the proposed model using three evaluation metrics, i.e., intersection over union (IoU), F_1 score, and overall accuracy (OA). IoU reflects the ability of the model to recognize foreground and background, F_1 is the comprehensive score of precision and recall, and OA evaluates the overall segmentation performance at the pixel level. The above three metrics are defined as follows:

$$\begin{aligned} \text{precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F_1 &= 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \end{aligned} \tag{9}$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{10}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{11}$$

where TP, TN, FP, and FN denote the number of true positives, true negatives, false positives, and false negatives, respectively. We calculate the F_1 score for each class and get the average of all classes to obtain the mean F_1 score (m-F_1) as the evaluation results.

C. Comparison With State-of-the-Art Methods

To verify the effectiveness of the proposed method, we compare it with other existing methods on two standard remote sensing image semantic segmentation datasets. The comparison methods include classical semantic segmentation methods, such as FCN [6], PSPNet [54], and DeepLabv3+ [29], and the state-of-the-art methods, such as S-RA-FCN [9], LANet [55], MFNet [56], SBANet [14], and MANet [41]. Meanwhile, several semantic segmentation methods based on mask classification, including MaskFormer [23] and Mask2Former [46], are also considered.

1) Experiments on Potsdam Dataset: The quantitative comparisons with other methods on the ISPRS Potsdam test set are shown in Table I. We can draw the following conclusions.

1) Our MMT has a significant improvement compared with other remote sensing image semantic segmentation methods and achieves state-of-the-art performance in mIoU, m-F_1, and OA.
2) In a horizontal comparison of the increment of our MMT relative to other methods, the increment of mIoU is the most significant of the three evaluation metrics. For example, compared to the classical method of FCN, MMT obtains increments of 7.26%, 5.02%, and 4.88% in mIoU, m-F_1, and OA, respectively, and the other methods show consistent results. As mentioned before, IoU reflects the ability of the model to recognize foreground and background, which indicates that the proposed MMA mechanism can effectively solve the problem of the foreground–background imbalance in remote sensing images.
3) For the F_1 score of each category, our MMT obtains more obvious improvements than MaskFormer (baseline) and achieves the best performance compared to other methods in the impervious surfaces, low vegetation, and tree classes. For the impervious surfaces, MaskFormer and its variants, and our MMT make prominent gains over other methods, which indicates that the mask classification has a notable advantage over per-pixel classification in learning the long-range spatial representation of background. However, the results of the car class in MMT are somewhat unsatisfactory, which is due to the fact that cars are discrete small targets, while the mask classification requires a large number of predicted masks for matching. Transformer models can effectively extract global context information, but their representations of local details are inferior to CNN models. The baseline MaskFormer is able to

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TABLE I
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS POTSDAM TEST SET

Method	Backbone	Imp. surf.	Building	Low veg.	Tree	Car	mIoU(%)	m-F_1(%)	OA(%)
FCN [6]	VGG-16	92.89	92.91	80.33	88.44	91.83	80.48	88.21	88.57
PSPNet [54]	ResNet-50	93.07	93.65	81.58	89.78	91.63	81.58	89.71	89.91
DeepLabv3+ [29]	ResNet-50	92.35	96.77	85.22	86.79	93.58	83.61	90.94	89.74
S-RA-FCN [9]	VGG-16	91.33	94.70	86.81	83.47	94.52	-	88.01	88.59
LANet [55]	ResNet-50	93.05	97.19	87.30	88.04	94.19	-	91.95	90.84
MFNet [56]	-	91.90	96.80	82.50	85.10	94.20	-	90.20	90.08
SBANet [14]	ResNet-101	93.83	**98.06**	88.97	89.48	**94.71**	-	93.01	92.80
MANet [41]	ResNet-50	93.40	96.96	88.32	89.36	96.48	86.95	92.90	91.32
MaskFormer [23]	Swin-B	96.44	96.78	86.98	89.89	83.93	85.46	91.78	91.62
Mask2Former [46]	Swin-B	97.96	96.92	88.44	90.66	84.59	86.61	92.54	92.52
MMT(Ours)	ResNet-50	98.03	96.68	88.37	92.43	86.85	86.58	92.61	92.78
MMT(Ours)	Swin-B	**99.83**	97.31	**89.69**	**93.60**	88.66	**87.74**	**93.23**	**93.45**

TABLE II
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE ISPRS VAIHINGEN TEST SET

Method	Backbone	Imp. surf.	Building	Low veg.	Tree	Car	mIoU(%)	m-F_1(%)	OA(%)
FCN [6]	VGG-16	88.17	92.99	81.54	83.62	85.42	74.22	84.92	86.95
PSPNet [54]	ResNet-50	88.04	94.17	80.82	84.57	82.88	74.88	85.38	87.19
DeepLabv3+ [29]	ResNet-50	91.35	94.34	81.32	87.84	78.14	76.31	86.60	88.91
S-RA-FCN [9]	VGG-16	91.47	94.97	80.63	88.57	87.05	-	88.54	89.23
LANet [55]	ResNet-50	92.41	94.90	82.89	88.92	81.31	-	88.09	89.83
MFNet [56]	-	92.20	**96.60**	81.70	89.10	89.70	-	89.90	90.20
SBANet [14]	ResNet-101	94.36	92.91	83.44	89.58	**91.43**	-	90.34	90.59
MANet [41]	ResNet-50	93.02	95.47	84.64	89.98	88.95	82.71	90.41	90.96
MaskFormer [23]	Swin-B	96.60	92.67	85.69	90.26	87.03	82.49	89.97	89.57
Mask2Former [46]	Swin-B	96.91	92.99	86.28	90.35	87.88	83.02	90.31	90.29
MMT(Ours)	ResNet-50	97.63	93.49	88.48	91.87	88.34	83.73	90.77	90.79
MMT(Ours)	Swin-B	**98.79**	94.23	**89.43**	**92.36**	88.64	**84.17**	**91.04**	**91.03**

achieve advanced levels in general but is far below other methods in the car. The proposed MMT can alleviate this problem by balancing foreground and background features and utilizing the MSL to enhance the representation of small targets, which outperforms the baseline by 4.73%. For the tree and low vegetation, they have high interclass similarity in color and texture, which is a challenge to correctly classify these two classes. Our MMT breaks the performance bottleneck and yields the result of 93.60% and 89.69% in the tree and low vegetation, respectively, depending on the proposed MMA mechanism, which can learn more definite interclass distance by enhancing the interdependent representations of background and foreground.

Qualitative comparisons on the ISPRS Potsdam test set are shown in Fig. 5. We select five representative local visualization examples to illustrate the superiority of the proposed method and add some dashed boxes as focus areas. In the first row of results, there is an area of low vegetation that is distinctly different in texture and color from the surrounding area, with large intraclass variability. The networks based on per-pixel classification, such as FCN and PSPNet, both seriously misidentify the low vegetation in this area, while our model can make accurate predictions. It indicates that the proposed MMT can adequately learn the interactive representation between foreground and background and also shows the importance of foreground–background balance in remote sensing image semantic segmentation, which is a crucial issue that previous methods have not solved. As shown in the lower right corner of the fourth row, there is a tree hidden among the low vegetation, barely perceptible even to the naked eye, but only MMT can extract the interclass differences and classify it correctly, which shows the effectiveness of the proposed method in dealing with high interclass similarity targets. Besides, the results of the remaining rows demonstrate that MMT is able to recognize the hard samples of buildings more precisely. In general, the proposed MMT is able to yield more accurate segmentation maps, especially for complex irregular targets, and obtains more refined boundary details. For the discrete small targets such as cars, benefiting from the MSL, there are also superior classification maps compared to the baseline.

2) Experiments on Vaihingen Dataset: Table II reports the quantitative comparisons on the ISPRS Vaihingen test set. The ISPRS Vaihingen dataset has fewer samples and lower spatial resolution than the ISPRS Potsdam dataset and greater

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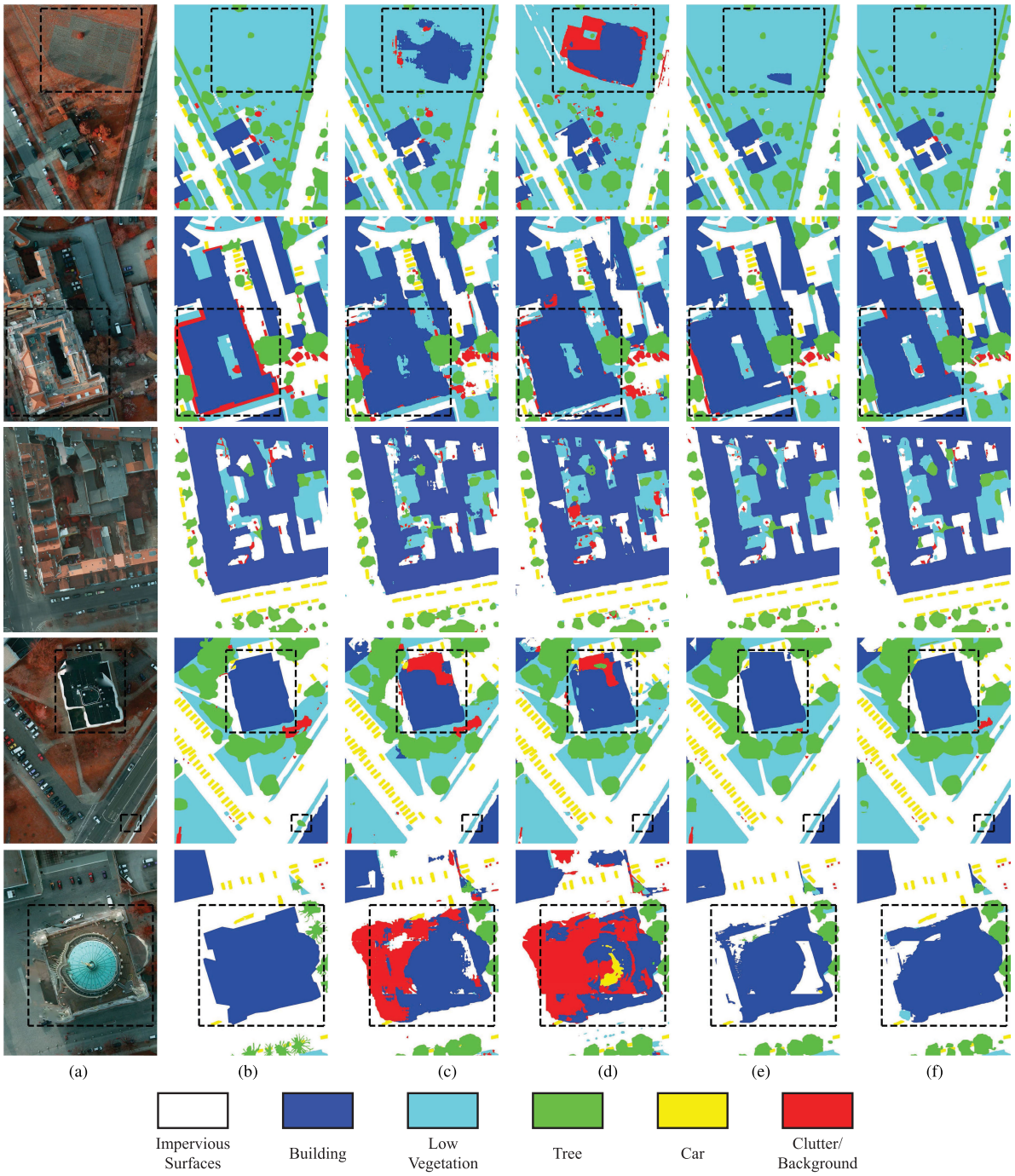


Fig. 5. Qualitative comparisons with different methods on the ISPRS Potsdam test set. (a) IRRG image. (b) Ground truth. (c) FCN. (d) PSPNet. (e) MaskFormer. (f) MMT(ours).

challenges for remote sensing image semantic segmentation. It is obvious that the following holds.

1) Our approach consistently outperforms other methods with significant margins in mIoU, m-F_1, and OA, which demonstrates the superiority and generalization capability of the proposed MMT.
2) Further analysis of the F_1 score of each category shows that the proposed MMT achieves the best performance in the three classes of impervious surfaces, low vegetation, and tree, and is second only to MFNet in the building class and to SBANet in the car class. It shows that our method is more capable of modeling complex irregular targets and further reveals the importance of mask classification-based methods for semantic segmentation of remote sensing images.
3) Compared to the baseline, the proposed MMT exceeds the MaskFormer by 1.68%, 1.07%, and 1.46% in mIoU, m-F_1, and OA, respectively, with the same backbone of Swin-B. It preliminary shows the advantages of the MMA mechanism and MSL, which can lead to a more precise and robust segmentation model, thus exploring a new paradigm for remote sensing image

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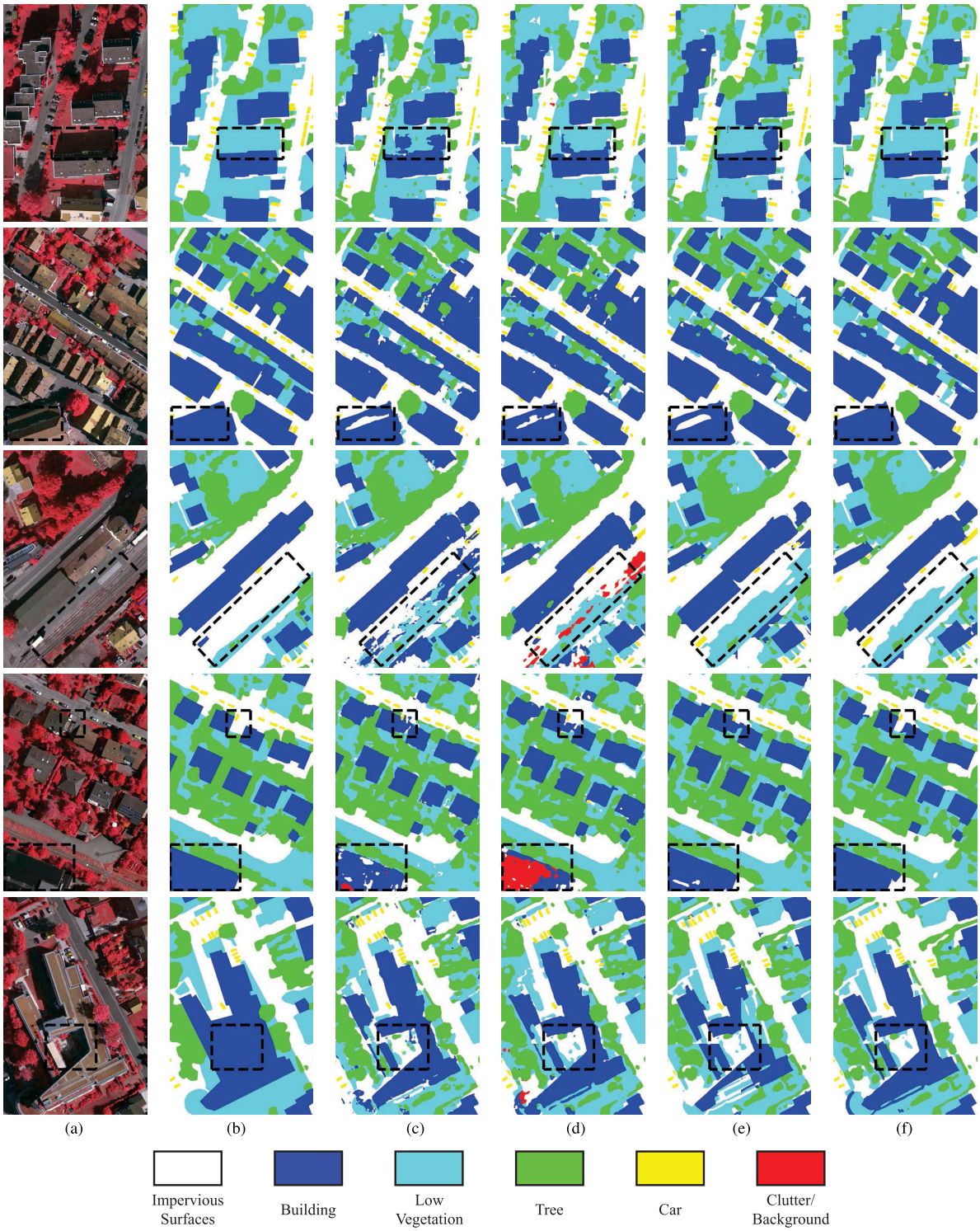


Fig. 6. Qualitative comparisons with different methods on the ISPRS Vaihingen test set. (a) RGB image. (b) Ground truth. (c) FCN. (d) PSPNet. (e) MaskFormer. (f) MMT(ours).

semantic segmentation with a foreground–background balance way. Further details of the ablation experiments are analyzed in Section IV-D.

Visualization results on the ISPRS Vaihingen test set are shown in Fig. 6, and we also select five typical examples for qualitative analysis. In the dashed box of the first row of results, there is a patch of low vegetation covered by the shadow of a building, causing it to be incorrectly identified as a building by all methods except MMT. Our method is able to make accurate predictions for such anomalous samples by learning global long-range interaction representations. The third row shows a region of impervious surfaces with complex intraclass variability. The compared networks based on per-pixel classification, such as FCN and PSPNet, both confusingly predict other classes, while our model makes relatively accurate predictions in this scene. It further demonstrates

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TABLE III
QUANTITATIVE COMPARISONS WITH THE STATE-OF-THE-ART METHODS ON THE LOVEDA DATASET

Method	Backbone	Per-class IoU							mIoU	m-F1	OA
		Background	Building	Road	Water	Barren	Forest	Agriculture			
FCN [6]	VGG-16	45.54	51.43	50.24	59.97	13.89	39.51	50.92	44.50	60.13	64.16
PSPNet [54]	ResNet-50	41.99	56.40	53.80	60.49	20.63	44.42	52.09	47.12	62.98	65.23
DeepLabv3+ [29]	ResNet-50	39.89	56.50	57.60	61.67	25.54	**47.87**	54.70	49.11	64.97	66.65
LANet [55]	ResNet-50	49.57	51.94	53.94	59.93	21.08	41.30	51.72	47.07	63.02	66.09
MaskFormer [23]	Swin-B	52.54	60.36	56.00	65.87	27.67	38.84	54.30	50.79	66.87	70.82
Mask2Former [46]	Swin-B	53.99	62.19	56.84	69.37	32.04	41.02	56.52	53.14	68.82	71.93
MMT(Ours)	ResNet-50	53.25	62.59	56.25	68.79	30.17	43.38	56.92	53.05	68.72	70.26
MMT(Ours)	Swin-B	**54.33**	**63.89**	**57.77**	**70.41**	**36.24**	44.58	**57.24**	**54.92**	**69.64**	**72.43**

the ability of our method in modeling contextual spatial correlation, which is the reason for the qualitative changes in the performance of the impervious surfaces class. As shown in the RGB image of the fifth row, the dashed box is, in fact, supposed to be the inner courtyard of a building, which belongs to the impervious surfaces class and has been pointed out for its incorrect annotations [56]. Compared with other methods, our proposed MMT is to present more realistic and meticulous segmentation results. In addition, the results of the second and fourth rows show the excellent performance of our method in the area of complex building clusters, almost close to the ground truth.

3) Experiments on LoveDA Dataset: The quantitative comparisons with other methods on the LoveDA dataset are shown in Table III. The LoveDA dataset has more and larger size images than ISPRS Potsdam and Vaihingen datasets. From the compared results, we can draw the following conclusions.

1) Our approach achieves the state-of-the-art performance of 54.92%, 69.64%, and 72.43% in mIoU, m-F_1, and OA, respectively, which proves that the proposed MMT also has excellent performance on large-scale remote sensing images.
2) For the IoU per class, our MMT obtains more obvious improvements than other methods in the background, building, road, water, barren, and agriculture classes. Especially in the barren class, there are large intra-class differences, which face challenges to its correct recognition. The proposed MMA mechanism can sufficiently learn the long-range contextual correlations of the foreground–background features and capture the key target representation, which improves by 10.48% compared with SANet [32].

Visualization comparisons on the LoveDA dataset are shown in Fig. 7. We select four typical local samples to illustrate the superiority of the proposed method. In the first row of results, there is a field being opened in the lower left corner, and there is a high interclass similarity between barren and agricultural classes. Among all methods, only our method is able to perform accurate classification, which demonstrates the effectiveness of the proposed MMT in dealing with high interclass similarity targets. As shown in the fourth row, the sample includes all seven classes, and the foreground and background features are complex. It can be seen that the visualization results of the proposed method are more accurate and can realize the land-cover segmentation of remote sensing images in complex environments. In addition, the results of the other rows show that the proposed MMT can also achieve accurate semantic segmentation for large-scale remote sensing images, which further proves the effectiveness of the proposed method.

In summary, through qualitative and quantitative comparisons, the proposed MMT achieves state-of-the-art performance in three evaluation metrics on the ISPRS Vaihingen and Potsdam 2-D datasets and the LoveDA land-cover dataset. In addition, for fair comparisons with the CNN-based methods, we further replace the backbone network with ResNet-50 in the experiments and achieve top-notch segmentation results on Potsdam, Vaihingen, and LoveDA datasets. The experimental results demonstrate the feasibility and effectiveness of redefining the semantic segmentation with mask classification pipeline for remote sensing image semantic segmentation. The mask classification separates foreground and background, which makes the network focus more on the recognition of the target mask but ignores the crucial background long-range interaction information in remote sensing images. The proposed approach improves the mask classification, which not only solves the problem of foreground–background imbalance but also introduces mask classification into the remote sensing image semantic segmentation in a real sense for the first time. We draw the following conclusions to explain the superiority of our MMT. First, different from the normal semantic segmentation methods that rely on the per-pixel classification to capture long-range information but suffer from the problem of foreground–background imbalance, we have explored a new paradigm based on mask classification for remote sensing image semantic segmentation. Second, the proposed MMA mechanism assists the network to learn more explicit intraclass interclass correlations by capturing context-interdependent representations with a foreground–background balance way. Finally, a novel MSL is proposed to efficiently introduce large-scale feature maps in Transformer while controlling the computational effort, so as to further improve the recognition performance of large-scale-varied targets in remote sensing images.

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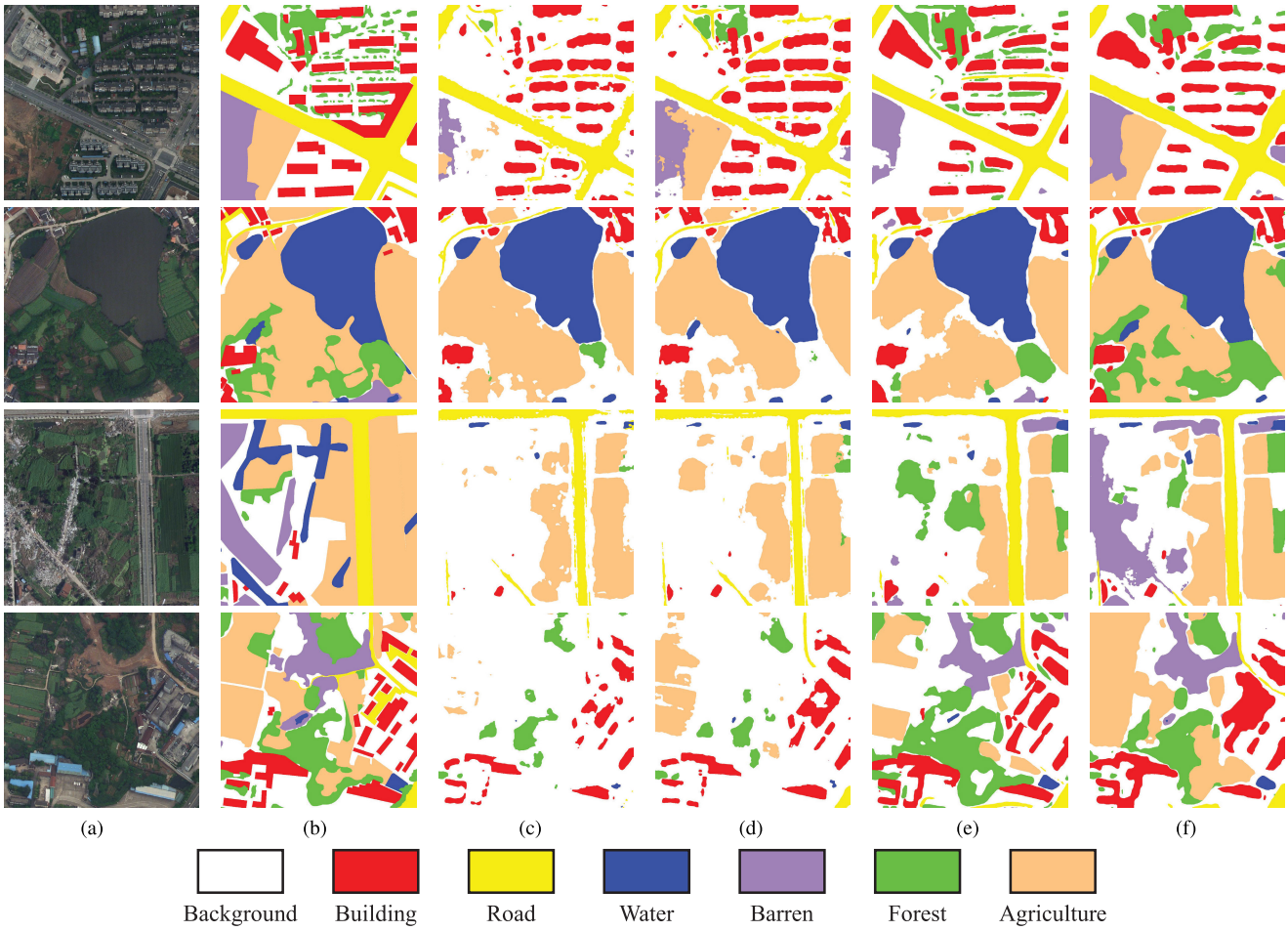


Fig. 7. Qualitative comparisons with different methods on the LoveDA dataset. (a) RGB image. (b) Ground truth. (c) FCN. (d) PSPNet. (e) MaskFormer. (f) MMT(ours).

TABLE IV
ABLATION STUDY OF THE NUMBER OF MASK QUERIES N_q ON THE ISPRS POTSDAM DATASET

N_q	mIoU (%)	m-F_1 (%)	OA (%)	FLOPs
10	86.63	92.74	92.88	131G
50	**87.74**	93.23	**93.45**	139G
100	87.72	**93.25**	93.43	150G
200	87.72	93.22	93.41	171G

D. Ablation Study

In this section, we further analyze the effectiveness of the proposed MMT from three aspects: hyperparameter setting, module performance, and computational complexity, respectively.

1) Study of Hyperparameters: In the proposed MTD, there are three hyperparameters that directly affect the performance of the network, which are the number of mask queries N_q, the size of the mask grid s_{grid} of the BGM, and the random ratio r_{random} of the FRM. For the number of mask queries N_q, it controls the representation ability of mask query features. According to the definition of mask classification, N_q should be greater than the number of classes, and the ablation study of the number of mask queries N_q on the ISPRS Potsdam dataset is shown in Table IV. It is obvious to conclude that the larger N_q is, the higher the computational complexity of the model is. When $N_q = 50$, the model achieves the best performance of 87.74% and 93.45% in mIoU and OA, respectively. As N_q gradually increases, the performance of our network tends to be stable. Therefore, considering the tradeoff between performance and computational complexity, N_q is set to 50.

For the mask grid s_{grid} of the BGM and the random ratio r_{random} of the FRM, they jointly influence the generation of the mixed-mask Mask_{mixed}, so we evaluate single one as a variable and keep the other to be the constant. We analyze the hyperparameter r_{random} with setting s_{grid} to 1/16 of the input feature map size, where r_{random} is ranging from 10% to 70%. OA is depicted in Fig. 8(a). It can be seen from the curve that OA increases first, then decreases, and yields the highest accuracy when r_{random} is equal to 50%. Similarly, when analyzing the hyperparameter s_{grid}, we set $r_{\text{random}} = 50\%$ and s_{grid} to 1/64, 1/32, 1/16, and 1/8 of the input feature size, respectively. It can be seen from Fig. 8(b) that OA performs the optimal when s_{grid} is equal to 1/16 of the input feature size. Therefore, hyperparameters s_{grid} and r_{random} are set to 1/16 and 50%, respectively.

2) Effectiveness Analysis of Modules: In the proposed MMT, the MMA mechanism focuses on solving the problem

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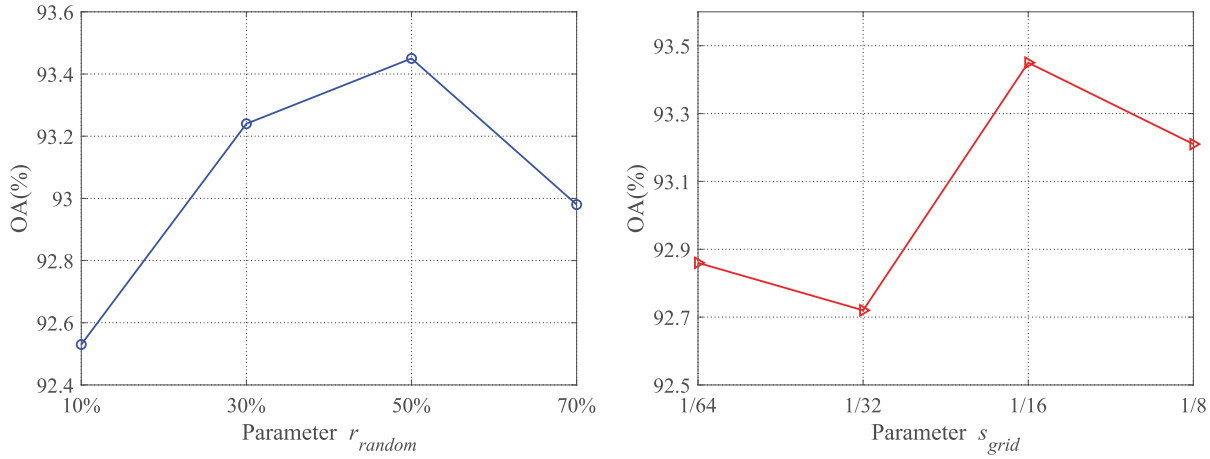


Fig. 8. Analysis of hyperparameters. (a) Hyperparameter r_{random}. (b) Hyperparameter s_{grid}.

TABLE V
ABLATION STUDY OF THE EFFECTIVENESS OF EACH MODULE ON THE ISPRS POTSDAM DATASET

Ablation settings	BGM	FRM	MSL	mIoU(%)	m-F_1(%)	OA(%)
1				85.46	91.78	91.62
2			✓	86.13(+0.68)	92.26(+0.48)	92.25(+0.63)
3	✓		✓	86.49(+1.03)	92.43(+0.65)	92.47(+0.85)
4		✓	✓	86.67(+1.21)	92.61(+0.83)	92.63(+1.01)
5	✓	✓		87.02(+1.56)	92.74(+0.96)	92.81(+1.18)
6	✓	✓	✓	87.74(+2.29)	93.23(+1.46)	93.45(+1.83)

TABLE VI
EFFICIENCY COMPARISONS OF DIFFERENT BACKBONE ON THE ISPRS POTSDAM DATASET

Method	Backbone	#params.			FLOPs			fps
		Backbone	Segmentation Head	Total	Backbone	Segmentation Head	Total	
Baseline	ResNet-50	23.46M	20.25M	43.71M	21.47G	44.10G	65.57G	27.2
	Swin-B	86.88M	20.25M	107.13M	93.59G	44.10G	137.68G	10.1
MMT	ResNet-50	23.46M	20.50M	43.95M	21.47G	45.24G	66.71G	26.5
	Swin-B	86.88M	20.50M	107.38M	93.59G	45.24G	138.83G	9.4

of foreground–background imbalance, and the MSL focuses on solving the problem of large-scale-varied targets in remote sensing images. To better evaluate the contribution of each module, we conduct ablation studies with six ablation settings to analyze the effectiveness of the proposed BGM, FRM, and MSL. Among the six ablation settings, the first one corresponds to the baseline MaskFormer, and the sixth one corresponds to the proposed MMT. The complete ablation experiments on the ISPRS Potsdam dataset are shown in Table V. It records how our network continuously improves segmentation performance, with the increase in relation to the baseline in brackets. We can draw the following conclusions.

1) When adding the MSL on the baseline, it shows a slight improvement in all metrics. Compared to the baseline with FPN [47], our MSL is able to integrate semantic and visual representation between large and small targets more efficiently.
2) The FRM can produce a more significant performance improvement than the BGM, which outperforms the baseline by 1.21%, 0.83%, and 1.01% in mIoU, m-F_1, and OA, respectively.
3) BGM, FRM, and MSL modules are nonconflicting, complementary, and mutually reinforcing, and they are able to promote the final model. Therefore, the proposed MMT can learn the interdependencies between foreground and background in a balanced way to achieve state-of-the-art performance on the Potsdam dataset.

3) Parameter Scales and Model Complexity: Moreover, Table VI studies the efficiency comparisons of different backbones on the ISPRS Potsdam dataset. For input data in the size of 512 × 512 pixels, we measure the computational

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efficiency with the number of parameters measured with million (M), floating-point operations per second (FLOPs), and frames/s (fps). A general semantic segmentation network consists of a backbone network and a segmentation head, and the improvement of our network relative to the baseline is mainly focused on the segmentation head, so it is essential to evaluate the efficiency of the backbone and the segmentation head separately. For the number of parameters, the number of the backbone network is the same for different methods under the same backbone, and the number of parameters of the segmentation head is the same for the same method under different backbones. Under the same backbone, the number of our model parameters is only 0.25 million higher than that of the baseline, while the performance exceeds the baseline by 1.56%, 0.96%, and 1.18% in mIoU, m-F_1, and OA. Similarly, with the same backbone, the FLOPs of our MMT are higher than the baseline by only 1.14 giga (G). It is obvious that the performance of the proposed method is significantly improved with only a few additional computations.

V. Conclusion

In this article, we introduce mask classification into remote sensing image interpretation for the first time and propose an MMT model for remote sensing image semantic segmentation within a foreground–background balance way. The experimental results on the ISPRS Vaihingen and Potsdam 2-D semantic labeling contest datasets demonstrate the superiority of the proposed approach compared with other related methods. The proposed MMA mechanism, a simple but effective module, assists the network to learn more explicit intraclass and interclass correlations, which is proven to solve the crucial problem of foreground–background imbalance. Moreover, an advanced MSL is proposed to integrate semantic and visual representations by efficiently utilizing large-scale feature maps in the Transformer, which is verified to solve the issue of large-scale-varied targets in remote sensing images.

Since semantic segmentation models yield weak performance on the small discrete targets, in the future, we will focus on the effective segmentation of small discrete targets in remote sensing images.

References

[1] Q. Zeng, J. Geng, W. Jiang, K. Huang, and Z. Wang, "IDLN: Iterative distribution learning network for few-shot remote sensing image scene classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[2] X. Chen, H. Ma, J. Wan, B. Li, and T. Xia, "Multi-view 3D object detection network for autonomous driving," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6526–6534.

[3] S. Chen, "SAR image speckle filtering with context covariance matrix formulation and similarity test," *IEEE Trans. Image Process.*, vol. 29, pp. 6641–6654, 2020.

[4] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 25, 2012, pp. 1–9.

[5] J. Geng, R. Wang, and W. Jiang, "Polarimetric SAR image classification based on feature enhanced superpixel hypergraph neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5237812.

[6] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[7] S. Chen and C. Tao, "PolSAR image classification using polarimetric-feature-driven deep convolutional neural network," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 627–631, Apr. 2018.

[8] G. Lin, A. Milan, C. Shen, and I. Reid, "RefineNet: Multi-path refinement networks for high-resolution semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5168–5177.

[9] L. Mou, Y. Hua, and X. X. Zhu, "Relation matters: Relational context-aware fully convolutional network for semantic segmentation of high-resolution aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 11, pp. 7557–7569, Nov. 2020.

[10] J. Geng, W. Jiang, and X. Deng, "Multi-scale deep feature learning network with bilateral filtering for SAR image classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 201–213, Sep. 2020.

[11] T. Panboonyuen, K. Jitkajornwanich, S. Lawawirojwong, P. Srestasathiern, and P. Vateekul, "Semantic segmentation on remotely sensed images using an enhanced global convolutional network with channel attention and domain specific transfer learning," *Remote Sens.*, vol. 11, no. 1, p. 83, 2019.

[12] J. Geng, X. Deng, X. Ma, and W. Jiang, "Transfer learning for SAR image classification via deep joint distribution adaptation networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5377–5392, Aug. 2020.

[13] S. Chen, X. Cui, X. Wang, and S. Xiao, "Speckle-free SAR image ship detection," *IEEE Trans. Image Process.*, vol. 30, pp. 5969–5983, 2021.

[14] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5400314.

[15] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021, pp. 1–22.

[16] Z. Xu, W. Zhang, T. Zhang, Z. Yang, and J. Li, "Efficient transformer for remote sensing image segmentation," *Remote Sens.*, vol. 13, no. 18, p. 3585, Sep. 2021.

[17] C. Zhang, W. Jiang, Y. Zhang, W. Wang, Q. Zhao, and C. Wang, "Transformer and CNN hybrid deep neural network for semantic segmentation of very-high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408820.

[18] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[19] Y. Li, S. Ouyang, and Y. Zhang, "Combining deep learning and ontology reasoning for remote sensing image semantic segmentation," *Knowl.-Based Syst.*, vol. 243, May 2022, Art. no. 108469.

[20] A. Ma, J. Wang, Y. Zhong, and Z. Zheng, "FactSeg: Foreground activation-driven small object semantic segmentation in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606216.

[21] X. Li et al., "PointFlow: Flowing semantics through points for aerial image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 4215–4224.

[22] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4095–4104.

[23] B. Cheng, A. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 17864–17875.

[24] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder–decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[25] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Cham, Switzerland: Springer, 2015, pp. 234–241.

[26] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," in *Proc. ICLR*, 2015, pp. 1–14.

[27] L. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[28] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 00:37:23 UTC from IEEE Xplore. Restrictions apply.

[29] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. ECCV*, Oct. 2018, pp. 833–851.
[30] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "Learning a discriminative feature network for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 1857–1866.
[31] Z. Huang et al., "CCNet: Criss-cross attention for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 6, pp. 6896–6908, Jun. 2023.
[32] Z. Zhong et al., "Squeeze-and-attention networks for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 13062–13071.
[33] B. Cheng et al., "SPGNet: Semantic prediction guidance for scene parsing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 5217–5227.
[34] R. Wang, P. Gong, and X. Liu, "Region-and-attention network for semantic segmentation," in *Proc. 7th Int. Conf. Comput. Commun. (ICCC)*, Dec. 2021, pp. 13927–13938.
[35] K. Nogueira, M. D. Mura, J. Chanussot, W. R. Schwartz, and J. A. Dos Santos, "Dynamic multicontext segmentation of remote sensing images based on convolutional networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7503–7520, Oct. 2019.
[36] G. Li, L. Li, H. Zhu, X. Liu, and L. Jiao, "Adaptive multiscale deep fusion residual network for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8506–8521, Nov. 2019.
[37] G. Wu et al., "Automatic building segmentation of aerial imagery using multi-constraint fully convolutional networks," *Remote Sens.*, vol. 10, no. 3, p. 407, Mar. 2018.
[38] Z. Xu, W. Zhang, T. Zhang, and J. Li, "HRCNet: High-resolution context extraction network for semantic segmentation of remote sensing images," *Remote Sens.*, vol. 13, no. 1, p. 71, Dec. 2020.
[39] R. Liu, L. Mi, and Z. Chen, "AFNet: Adaptive fusion network for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7871–7886, Sep. 2021.
[40] X. Yang et al., "An attention-fused network for semantic segmentation of very-high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 177, pp. 238–262, Jul. 2021.
[41] R. Li et al., "Multiattention network for semantic segmentation of fine-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4401120.
[42] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4401120.
[43] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
[44] X. Wang, T. Kong, C. Shen, Y. Jiang, and L. Li, "SOLO: Segmenting objects by locations," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 649–665.
[45] N. Carion, F. Massa, G. Synnaeve, N. Usunier, A. Kirillov, and S. Zagoruyko, "End-to-end object detection with transformers," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 213–229.
[46] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1280–1289.
[47] T. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 936–944.
[48] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
[49] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.
[50] ISPRS. *2D Semantic Labeling Contest*. Accessed: Jan. 10, 2023. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/semantic-labeling.aspx
[51] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.
[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[53] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. ICLR*, 2019, pp. 1–19.
[54] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.
[55] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.
[56] Y. Sun, Z. Fu, C. Sun, Y. Hu, and S. Zhang, "Deep multimodal fusion network for semantic segmentation using remote sensing image and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5404418.



Zhe Xu received the B.S. degree in electronic science and technology and the master's degree in electronic information from Northwestern Polytechnical University, Xi'an, China, in 2020 and 2023, respectively.

His research interests include remote sensing image processing and deep learning.



Jie Geng (Member, IEEE) received the B.S. degree in electronic and information engineering from the Dalian University of Technology, Dalian, China, in 2013, and the Ph.D. degree from the School of Information and Communication Engineering, Dalian University of Technology, in 2018.

He is currently an Associate Professor with the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China. His research interests include synthetic aperture radar (SAR) image processing, remote sensing image classification, and few-shot learning.



Wen Jiang received the bachelor's and master's degrees from PLA Information Engineering University, Zhengzhou, China, in 1994 and 1997, respectively, and the Ph.D. degree from Northwestern Polytechnical University, Xi'an, China, in 2009.

She is currently a Professor with the School of Electronics and Information, Northwestern Polytechnical University. Her research interests are in the areas of information fusion and artificial intelligence with uncertainty.

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